Phase 6: Project Testing

1. Testing Strategy

Ensuring the reliability of the **Rising Waters** prediction system involved multiple layers of testing, from data validation to endpoint verification.

2. Model Evaluation (Data Science Testing)

The Machine Learning models were tested using a standard 80/20 train-test split. The evaluation metrics used were Accuracy, Precision, Recall, and the F1-Score.

- **Decision Tree:** ~85% Accuracy
- **Random Forest:** ~91% Accuracy
- **K-Nearest Neighbors (KNN):** ~88% Accuracy
- **XGBoost: 96.55% Accuracy** (Selected for Production)

The models were tested against edge cases (e.g., extremely high annual rainfall with low cloud visibility) to ensure the predictions logically aligned with meteorological expectations.

3. Application Testing (Software Engineering)

- **Unit Testing:** Individual functions inside `train_model.py` and data transformation logic within the Flask routes were tested to ensure they processed input arrays correctly.
- **Integration Testing:** Verifying that the Flask application successfully deserialized the `.pkl` model and that the HTML form data successfully mapped to the model's expected feature array.
- **UI/UX Testing:** The web interface was tested across different screen sizes to ensure responsive design CSS operated correctly. Form validation was tested to prevent users from submitting text instead of numeric data.